Enrico Pallotta

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Education

University of Bonn Mar 2024 - Mar 2028
Ph.D. in Computer Vision *Bonn, Germany*

– Supervisor: Professor Dr. Juergen Gall.

– Topics: Video Diffusion models, Controllable Video Generation, World Models.

University of Bologna Sep 2021 - Oct 2023
MSc in Artificial Intelligence *Bologna, Italy*

– Core courses: *Image Processing and Computer Vision, Machine Learning for Computer Vision, Natural Language Processing, Deep Learning, Combinatorial Decision Making and Optimization.*

– Final grade: 110/110 summa cum laude.

University of Bari Sep 2018 - Jul 2021
BSc in Computer Science *Bari, Italy*

– Thesis: *"Segmentation of lung nodules with Mask R-CNN".*

– Final grade: 110/110 summa cum laude.

Work Experience

Amazon Sep 2026 - Dec 2026
Applied Scientist Intern *Berlin, Germany*

– Working on multimodal alignment in self-supervised learning.

Technology Innovation Institute Feb 2023 - Sep 2023
Research Intern *Abu Dhabi, UAE*

– Perception and computer vision research intern in the autonomous robotics research center for the development of the perception module for racing drones, but also trajectory generation for control, sensor fusion algorithm, deep learning model optimization through CUDA and TensorRT.

– Working on my master thesis and a RA-L paper concerning the collection and annotation of an extended dataset for drone racing and the implementation of both deep learning and computer vision techniques for objects pose estimation.

Publications

1. ***Enrico Pallotta**, *Sina Raoufi, Lars Doorenbos, Gianni Franchi, Juergen Gall. "Generative Uncertainty as a Self-supervised Signal for Semantic Similarity Learning". Under review, preprint arXiV:2609.35341.

2. M. Halstead, E. Guclu, M. Farag, **E. Pallotta**, C. Hund, R. Roscher, M. Bennewitz, J. Gall, C. Stachniss, C. McCool. Still image and spatial-temporal tomato data enabling detection, segmentation, tracking, and video-instance segmentation using strong and weak labels. Under review, preprint arXiv:2607.14934

3. Sami Azirar, **Enrico Pallotta**, Jan Nogga, Juergen Gall, Sven Behnke, Hermann Blum. "ContactFlow: A video action conditioning that transfers across embodiments". Under review, preprint arXiv:2607.26579.

4. ***Enrico Pallotta**, *Sina Mokhtarzadeh Azar, Lars Doorenbos, Serdar Ozsoy, Umar Iqbal, Juergen Gall. "EgoControl: Controllable Egocentric Video Generation via 3D Full-Body Poses", IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**) 2026.

5. **Enrico Pallotta**, Sina Mokhtarzadeh Azar, Shuai Li, Olga Zatsarynna, Juergen Gall. "SyncVP: Joint Diffusion for Synchronous Multi-Modal Video Prediction", IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**) 2025.

6. Sina Mokhtarzadeh Azar, Emad Bahrami, **Enrico Pallotta**, Gianpiero Francesca, Radu Timofte, Juergen Gall. "Sequence-Adaptive Video Prediction in Continuous Streams using Diffusion Noise Optimization", Computer Vision and Image Understanding (**CVIU**), 2026.

7. M. Bosello, D. Aguiari, Y. Keuter, **E. Pallotta**, S. Kiade, G. Caminati, F. Pinzarrone, J. Halepota, J. Panerati, G. Pau. "Race Against the Machine: a Fully-annotated, Open-design Dataset of Autonomous and Piloted High-speed Flight", IEEE Robotics and Automation Letters (**RA-L**) 2024.

8. *Yuri Noviello, ***Enrico Pallotta**, *Flavio Pinzarrone, *Giuseppe Tanzi, "TeamUnibo at SemEval-2023 Task 6: A transform based approach to Rhetorical Roles prediction and NER in Legal Texts" , Proceedings of the 17th International Workshop on Semantic Evaluation (**SemEval**) 2023.

Skills

Research:	World Models, Multi-modal Video Generation, World Action Models, Video Forecasting, Vision-Language Models, Visual Foundation Models.
Programming:	Python (Advanced), C/C++ (Medium), CUDA, Java, javascript, PHP, HTML.
Tools:	PyTorch, Torch Parallel (DDP, FSDP), AI coding agents, Linux, ROS, TensorRT.
Languages:	English (full proficiency), Italian (Native), German (A1).

Awards and Grants

1. **Top reviewer** (NeurIPS 2026)

2. **Outstanding Paper Award**, ECCV 2026 Workshop *Human Motion-Informed World Models and Socially Intelligent Action* (HuMoWM) for the paper *EgoControl*.

3. **Outstanding Paper Award**, ECCV 2026 Workshop *Universal Representations for Perception, Reasoning, and World Modeling* (UniWorld) for the paper *EgoControl*.

4. **Compute award (Jupiter SuperComputer)**: 750k GH200 GPU hours for Holistic Multi-modal Egocentric Video Forecasting: working on egocentric controllable video generation for empowering Embodied AI.

Academic activities

- **Conference/Journal Reviewer**: CVPR (2025, 2026), NeurIPS (2026).
- **Teaching assistant**: Video Analytics, Computer Vision, Computer Vision Lab and Seminar.